

# Haeyun Choi

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## Research Interests

My research interests span various topics in computer vision, emphasizing real-world applications. I am currently focused on robust 3D scene representations and rendering techniques that remain effective under challenging conditions. The goal of my research is to move beyond idealized assumptions, embracing the complexities of real-world scenarios.

## Education

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|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>MS</b> | <b>POSTECH</b> , Artificial Intelligence                                                                                                                            | Mar 2023 – Feb 2025 |
|           | <ul style="list-style-type: none"> <li>Advisor: <a href="#">Prof. Sunghyun Cho</a></li> <li>Thesis: "Exploiting Deblurring Networks for Radiance Fields"</li> </ul> |                     |
| <b>BS</b> | <b>Korea University</b> , Biomedical Engineering                                                                                                                    | Mar 2015 – Aug 2022 |

## Publications

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Exploiting Deblurring Networks for Radiance Fields</b>                                                                                                                                                            | 2025 |
| <b>Haeyun Choi</b> , Heemin Yang, Janghyuk Han, Sunghyun Cho<br>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2025)<br>Workshop on Image Processing and Image Understanding (IPIU 2025, oral) |      |
| <b>Optical Flow-based Shakiness Detection in Dash-cams for Hit-and-Run Accident Analysis</b>                                                                                                                         | 2024 |
| <b>Haeyun Choi</b> , Sunghyun Cho<br>Workshop on Image Processing and Image Understanding (IPIU 2024)                                                                                                                |      |
| <b>AutoCycle-VC: Towards Bottleneck-Independent Zero-Shot Cross-Lingual Voice Conversion</b>                                                                                                                         | 2023 |
| <b>Haeyun Choi</b> , Jio Gim, Yuho Lee, Youngin Kim, Young-Joo Suh<br>arXiv preprint                                                                                                                                 |      |

## Awards and Honors

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|--------------------------------------------------------------------------------------|-------------|
| <b>Outstanding Paper Award, IPIU 2025</b> ( <a href="#">link</a> )                   | 2025        |
| <b>KT-POSTECH AI Expert Program Scholarship</b> \$7,800 x 4 ( <a href="#">link</a> ) | 2023 - 2025 |
| <b>Excellence Prize, POSCO Youth Academy AI Competition</b>                          | 2022        |
| <b>Grand Prize, AI for Health &amp; SDGs Competition</b> ( <a href="#">link</a> )    | 2021        |
| <b>Hongsan Scholarships</b> \$1,700                                                  | 2021        |
| <b>Mirae Asset Exchange Student Scholarships</b> \$3,100 ( <a href="#">link</a> )    | 2021        |

## Experience

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>KT R&amp;D Center</b> , AI Research Scientist                                                                                                          | Seoul, South Korea  |
| <ul style="list-style-type: none"> <li>Curating multimodal datasets for KT's proprietary MLLM</li> </ul>                                                  | Mar 2025 – Present  |
| <b>POSTECH Institute of Artificial Intelligence</b> , Research Scientist                                                                                  | Pohang, South Korea |
| <ul style="list-style-type: none"> <li>PI : <a href="#">Prof. Young-Joo Suh</a></li> <li>Voice conversion using mel-spectrogram representation</li> </ul> | May 2022 – Feb 2023 |

## Projects

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### Real-time Danger Alert for Convenience Stores by Motion and Voice Recognition

2022

- Designed a real-time violence detection system for store clerks
- Used top-down pose estimation and lightweight classification models
- Built a 20k image dataset and enabled SMS alerts with incident frames

### Wear Your Mask On: Real-time Mask Detection & Warning System

2021

- Built a real-time mask alert system for COVID-19, supporting public safety
- Enabled real-time video inference using MobileNet and Haar Cascade

## Skills & Languages

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**Skills:** Python, PyTorch, Blender, Matlab

**Languages:** Korean (Native), English (Proficient)

## Extracurricular Activities

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### Team-Mate Mentorship Program (K-Digital Training [🔗](#), KCCI [🔗](#))

Aug 2024 – Present

- Selected as **1st cohort mentor** [🔗](#) in a national career development program
- Peer mentorship program, supporting junior trainees in career navigation
- Engaged in **job camps** [🔗](#) and **social contribution programs** [🔗](#)

### Military Service (Republic of Korea Air Force [🔗](#))

Aug 2016 – Aug 2018

- Sergeant as Operations
- Maintained 24/7 emergency standby status for mission deployment
- Managed mission planning and scheduling for search and rescue operations

### Student Representative (Biomedical Engineering, Korea University)

Mar 2015 – Aug 2016

- Volunteered as student representative
- Helped organize departmental events and student activities